

Ratio and Proportion — Paper A

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- To share a total: **add the parts, divide the total by that sum, then multiply back up.**
- For proportion questions, **find one first, then scale.**
- Give every ratio in its simplest form, and convert to a common unit before simplifying.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – WRITING AND SIMPLIFYING RATIOS

1. Complete the table. Each empty box is worth one mark. [6]

Ratio	Simplest form	First share as a fraction of the total
18 : 24		
20 : 50		
45 : 27		

The last column is where the marks are lost. Add both numbers of the ratio to get the total number of parts first.

2. Write each ratio in its simplest form.

(a) 24 : 36 [1]

(b) 5 : 15 : 20 [1]

3. Write each of these as a ratio in its simplest form. Convert the units first.

(a) 30 cm : 1 m [1]

(b) 45 minutes : 2 hours

[1]

SECTION 2 – SHARING A QUANTITY

4. Share 350 in the ratio 2 : 5.

[2]

5. Share 720 in the ratio 4 : 5.

[2]

6. Share ₹960 among three people in the ratio 1 : 2 : 5.

[3]

Check your three answers add back up to 960 before you move on.

7. In a class the ratio of boys to girls is 3 : 4. There are 28 students in the class.

(a) How many boys are there?

[1]

(b) How many girls are there?

[1]

- 8.** A recipe for 4 people uses 300 g of rice. How much rice is needed for 10 people? [2]

- 9.** 5 pens cost ₹60. Find the cost of 8 pens. [2]

- 10.** Concrete is made by mixing cement and sand in the ratio 1 : 4. A builder uses 60 kg of cement.

- (a)** What mass of sand does he use? [1]

- (b)** What is the total mass of the concrete he makes? [1]

- 11.** The ratio of red to blue counters in a bag is 2 : 3. There are 18 red counters.

- (a)** How many blue counters are there? [1]

- (b)** How many counters are there altogether? [1]

12. Two shops sell the same rice.

[3]

Shop A: 2 kg for ₹150. **Shop B:** 5 kg for ₹365.

Work out the price per kilogram at each shop, and state which is better value.

Two correct prices and no sentence saying which shop wins scores 2 out of 3.

End of paper. Check every answer has a unit or a form the question actually asked for.